LAB 1 – Deploying Autonomous AI Agents via Azure AI Foundry Portal

Project Goal

Access to a pre-provisioned Azure AI Foundry Project workspace, confirm model deployment health, and build an autonomous AI Agent with Code Interpreter tool capabilities using the Azure AI Foundry Web Portal.

Architecture Overview

This lab simulates a corporate training workflow where cloud engineers access a pre-created AI Project containing deployed foundation models, configure an autonomous agent, enable the Code Interpreter tool, and validate its output in the Agent Playground.

Technologies Used

• Microsoft Foundry Web Portal (ai.azure.com)

• Model Catalog (Pre-deployed gpt-5.4-mini)

• Agent Playground

• Pre-built Code Interpreter Tool

LAB IMPLEMENTATION

Step 1 — Open Web Browser

Execution

Open Google Chrome/ Microsoft Edge, or any supported web browser.

Expected Output / Outcome

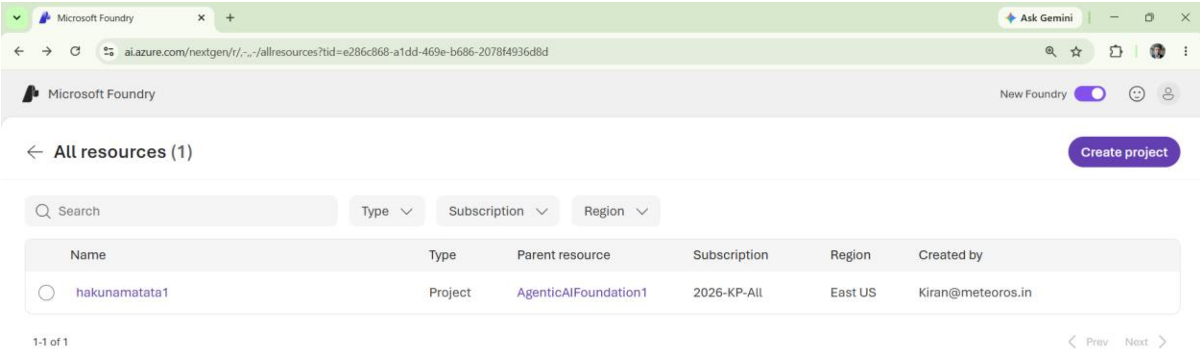
The browser opens and is ready to use.

Step 2 - Open Azure AI Foundry

Execution

Navigate to the Azure AI Foundry portal using the following URL:

https://ai.azure.com



Expected Output / Outcome

The Azure AI Foundry portal opens and displays the available projects or prompts the user to sign in.

Step 3 - Open the Lab Project

Execution

Click the Azure AI Foundry project provided by the lab team (for example: hakunamatata1).

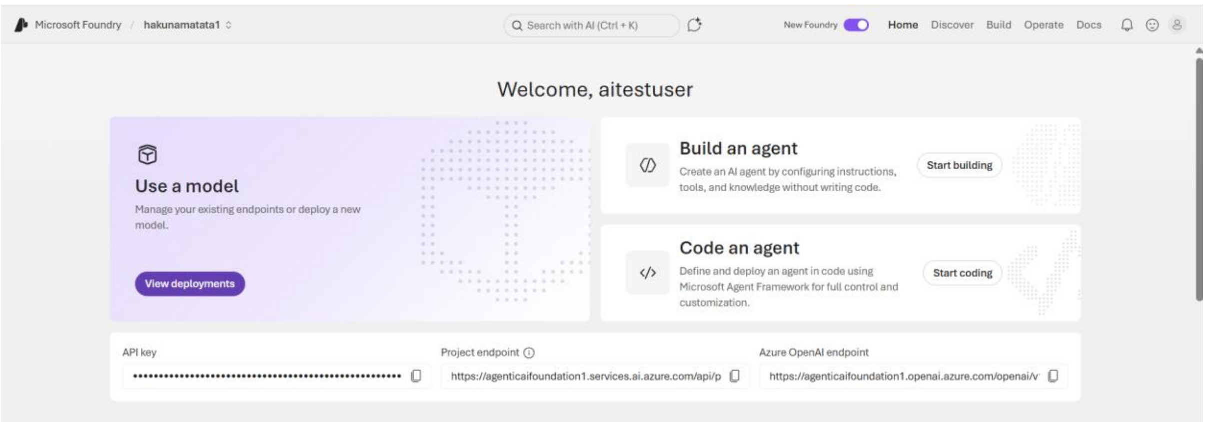
Expected Output / Outcome

The project workspace opens and displays the project dashboard, navigation menu, and available resources.

Step 4 - Verify Project Access

Execution

Verify that the project home page opens successfully and displays the project resources, endpoints, and available AI development options.



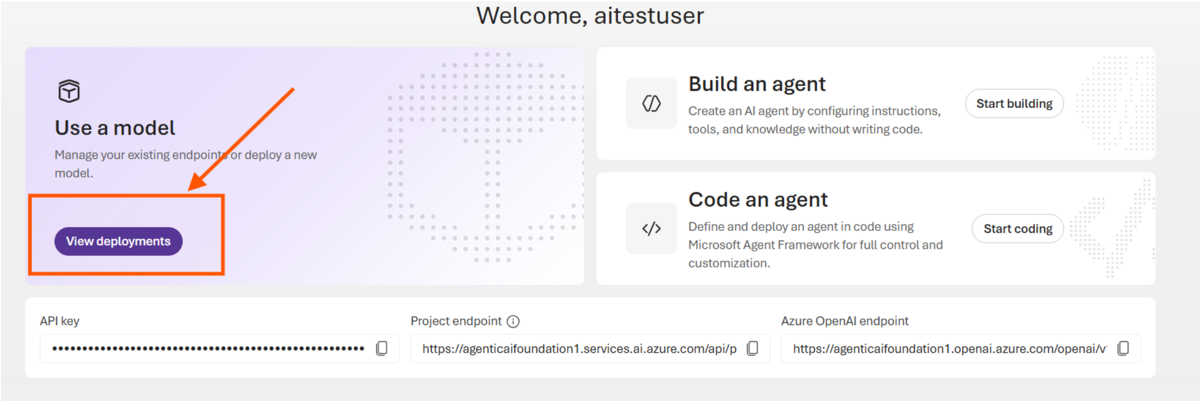
Expected Output / Outcome

The project dashboard is displayed, showing the API key, project endpoint, Azure OpenAI endpoint, and agent development options.

Step 5 - Open Deployments

Execution

On the project home page, click **View deployments**.



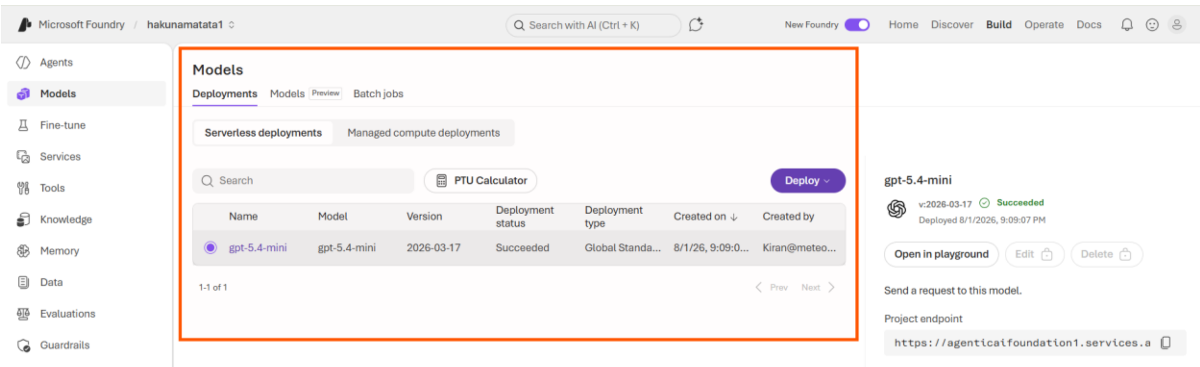
Expected Output / Outcome

The Deployments page opens and displays the available model deployments for the project.

Step 6 - Verify Model Deployment Status

Execution

Locate the **gpt-5.4-mini** deployment in the deployments list and verify that its deployment status is **Succeeded**.



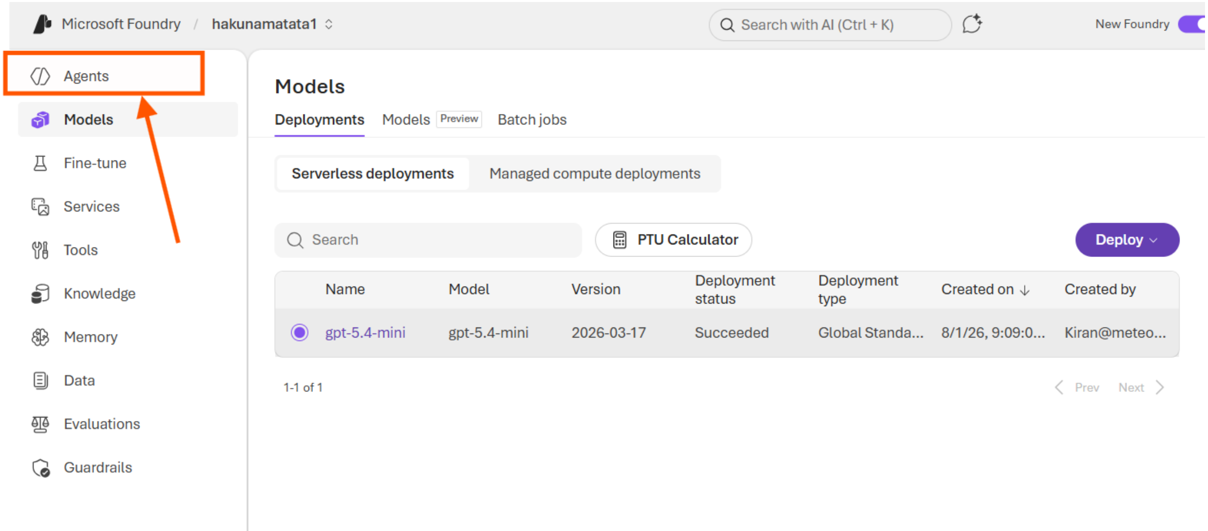
Expected Output / Outcome

The **gpt-5.4-mini** deployment is visible and shows a **Succeeded** status, confirming that the model is available for use.

Step 7 - Open the Agents Page

Execution

In the left navigation menu, click **Agents**.



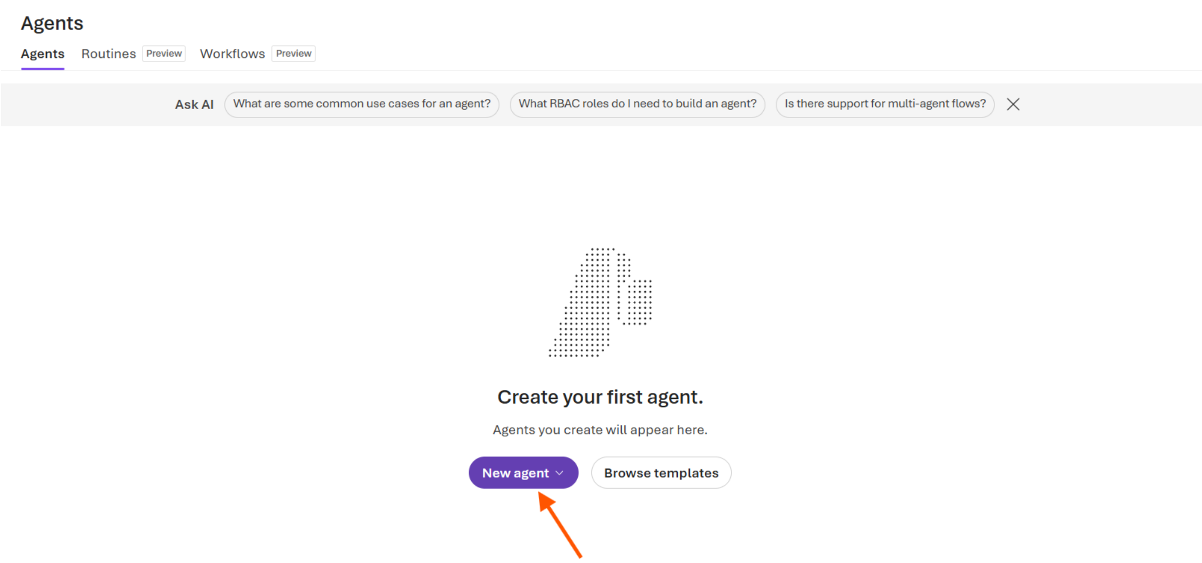
Expected Output / Outcome

The Agents page opens and displays the available agents. If no agents have been created, the page displays a **Create your first agent** message along with the **New agent** button.

Step 8 - Create a New Agent

Execution

Click **New agent** on the Agents page.



Expected Output / Outcome

A dropdown menu opens and displays the available agent creation options:

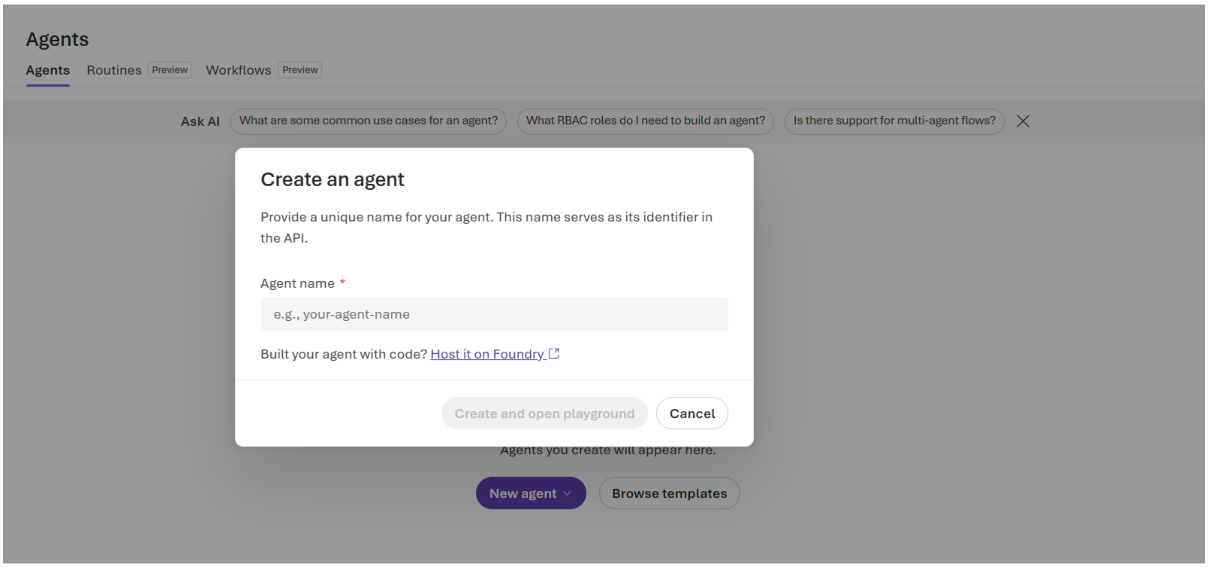
- **Build an agent**
- **Code an agent**
- **Link external agent**

For this lab, select **Build an agent** to continue with agent creation.

Step 9 - Select Build an Agent

Execution

From the **New agent** dropdown menu, select **Build an agent**.



Expected Output / Outcome

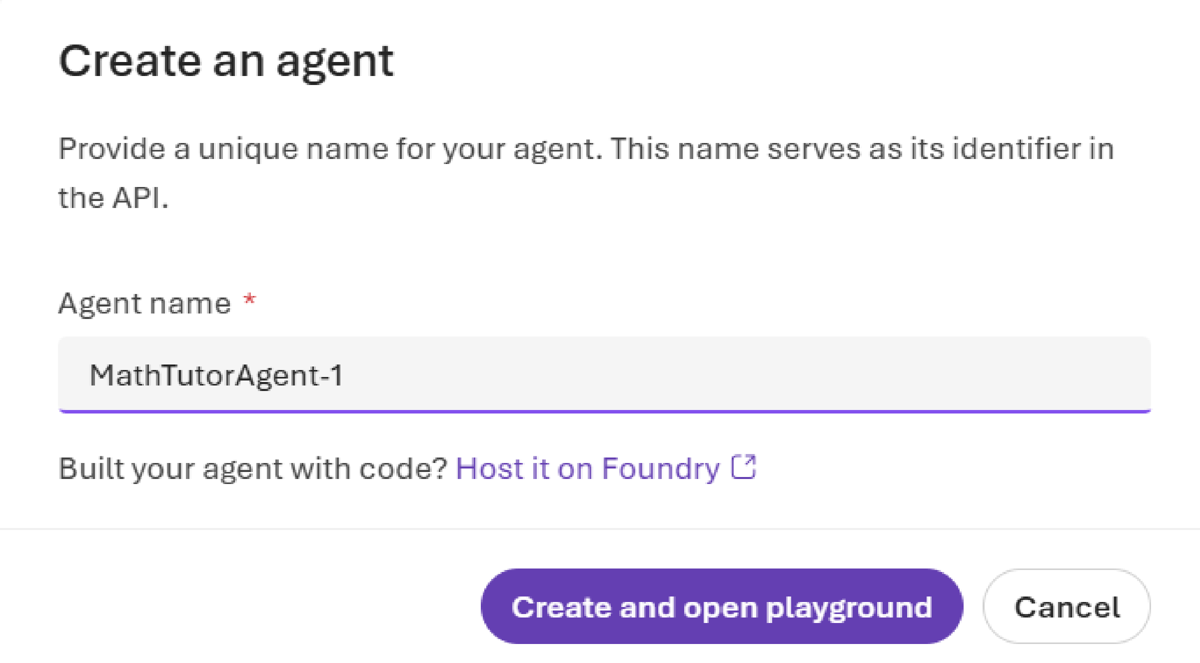
The **Create an agent** dialog box opens and displays the **Agent name** field along with the **Create and open playground** option.

Step 10 - Set Agent Name

Execution

In the **Agent name** field, enter:

MathTutorAgent-1



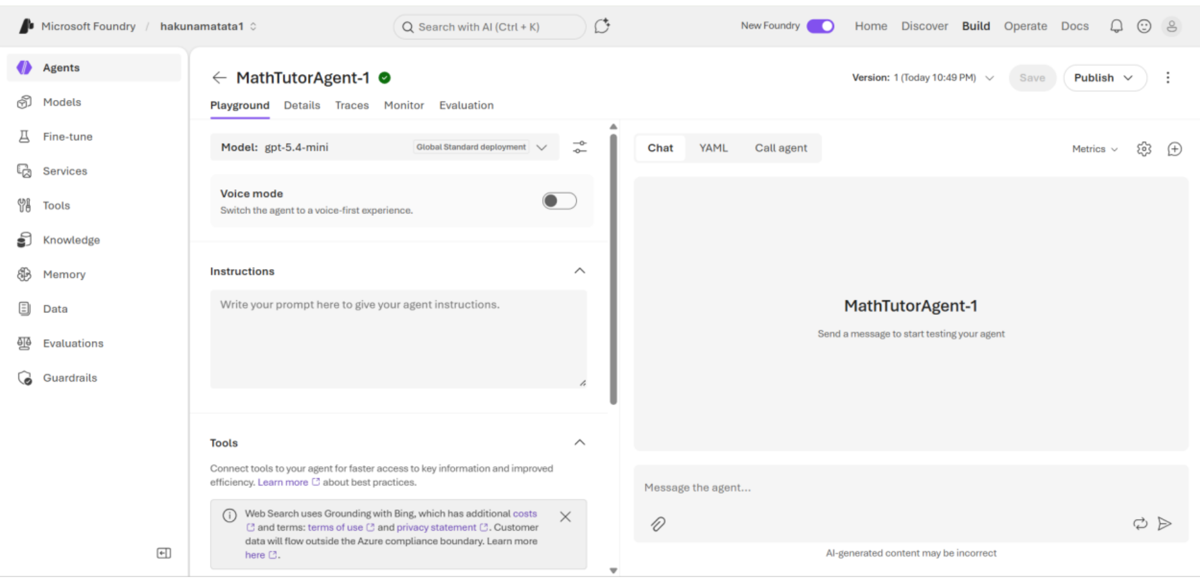
Expected Output / Outcome

The agent name field is populated with **MathTutorAgent-1**, and the **Create and open playground** button becomes available.

Step 11 - Open the Agent Playground

Execution

Click **Create and open playground** after entering the agent name.



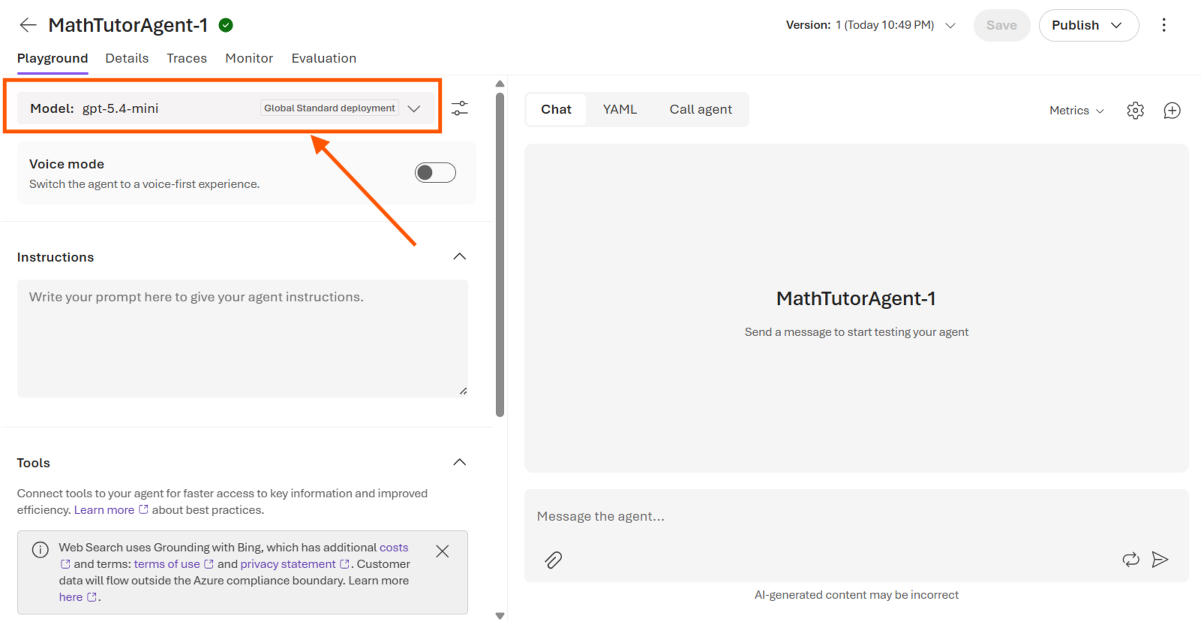
Expected Output / Outcome

The agent playground opens and displays the newly created agent. The page shows the selected model, instructions section, tools section, and an interactive chat interface for testing the agent.

Step 12 - Verify the Selected Model

Execution

Verify that **gpt-5.4-mini** is selected in the **Model** field at the top of the playground.



Expected Output / Outcome

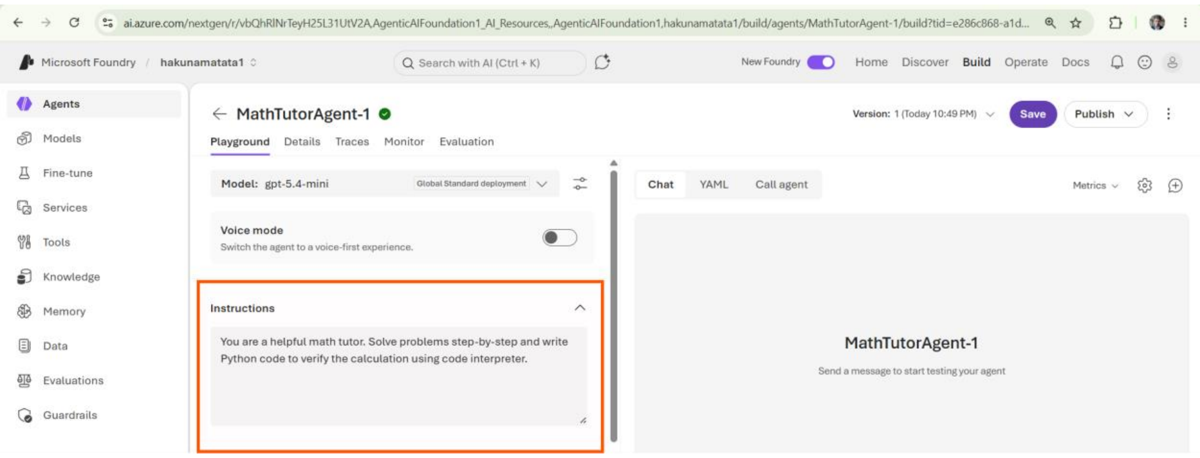
The **gpt-5.4-mini** model is displayed as the active model for the agent.

Step 13 - Configure Agent Instructions

Execution

In the **Instructions** section, enter the following prompt:

You are a helpful math tutor. Solve problems step-by-step and write Python code to verify the calculation using code interpreter.



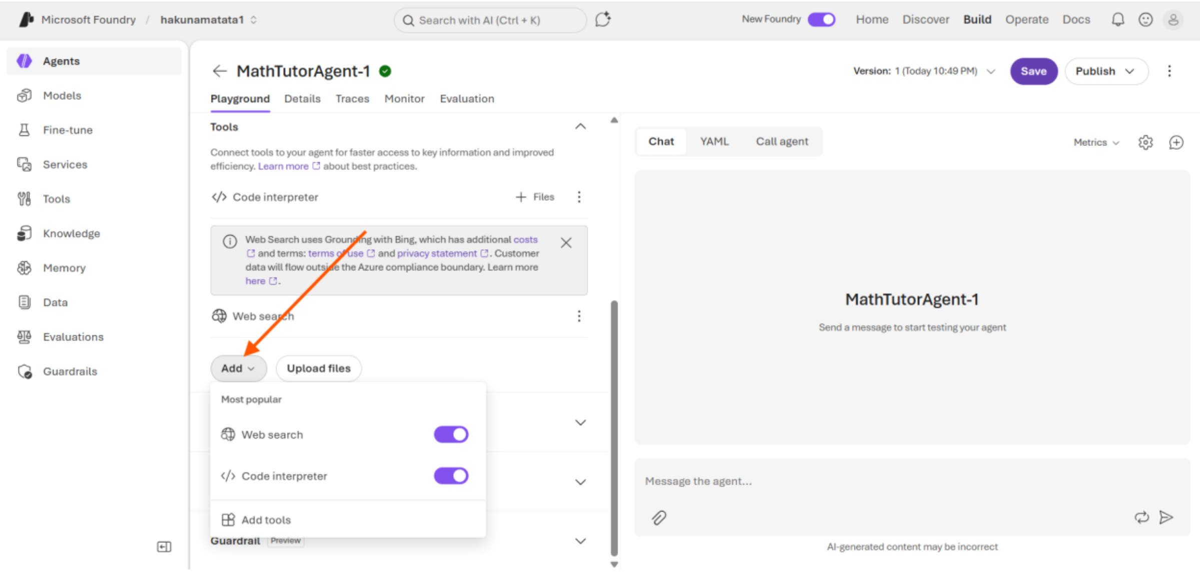
Expected Output / Outcome

The instructions are added to the agent configuration and are displayed in the Instructions section.

Step 14 - Enable Code Interpreter Tool

Execution

In the **Tools** section, click **Add** and enable the **Code Interpreter** tool.



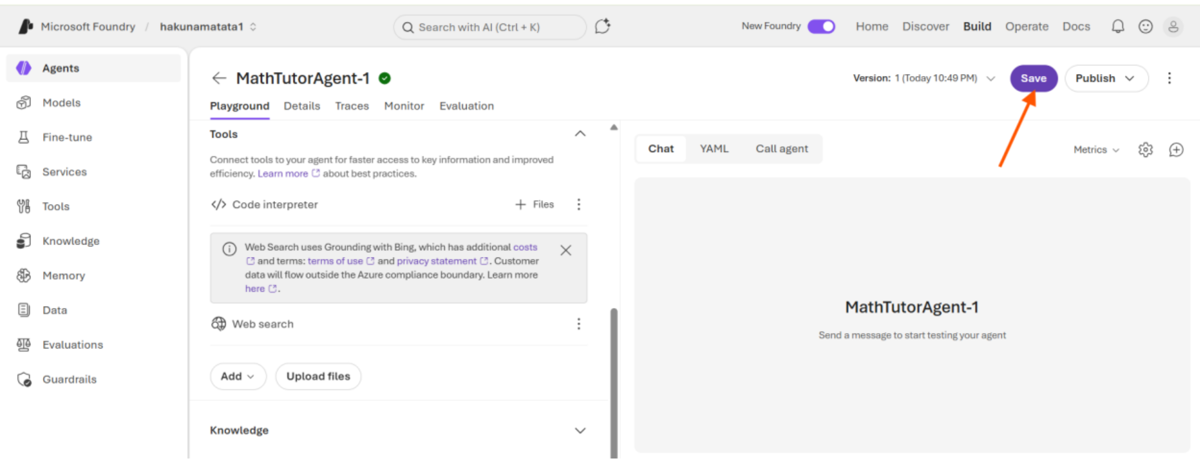
Expected Output / Outcome

The **Code Interpreter** tool is enabled and added to the agent, allowing the agent to execute Python code during conversations.

Step 15 - Save Agent Configuration

Execution

Click **Save** in the top-right corner of the page.



Expected Output / Outcome

The agent configuration is saved successfully, and the latest version of the agent is displayed.

Step 16 - Open the Agent Playground

Execution

After saving the agent configuration, remain on the **Playground** tab.

Expected Output / Outcome

The Playground tab is displayed and shows the agent configuration, available tools, chat interface, and message input box for testing the agent.

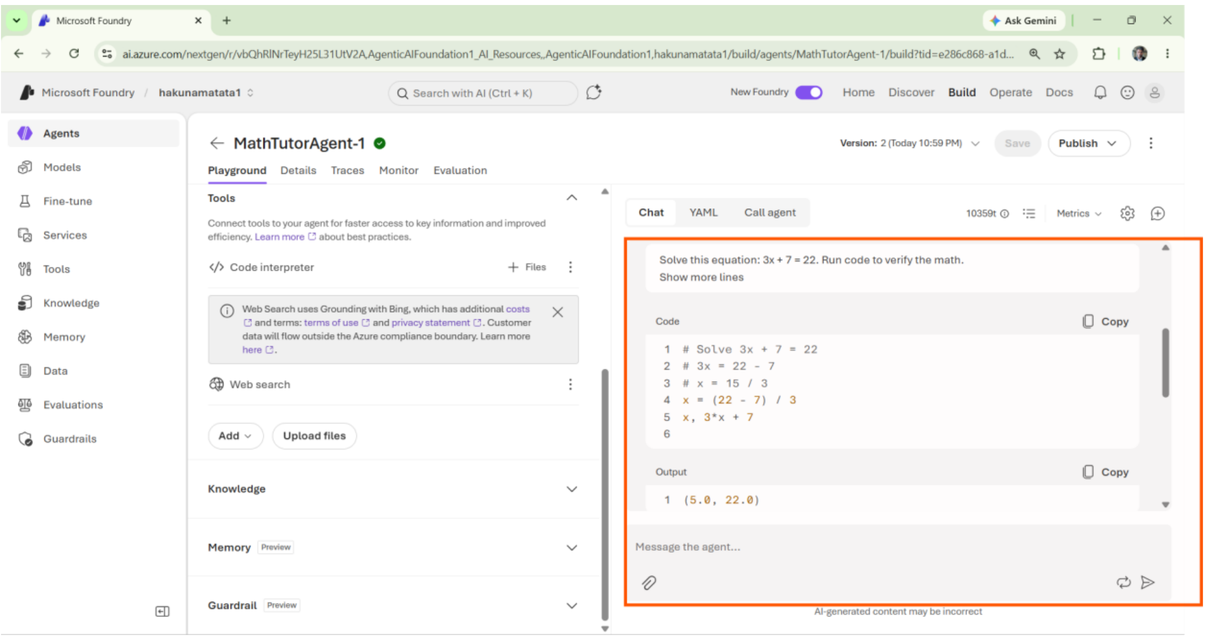
Step 17 - Test the Agent in the Playground

Execution

In the message box, enter the following prompt:

Solve this equation: 3x + 7 = 22. Run code to verify the math.

Click **Send**.



Expected Output / Outcome

The agent processes the request, uses the **Code Interpreter** tool to execute Python code, and returns the correct solution. The response confirms that:

x = 5

along with the Python verification showing:

3 * 5.0 + 7 = 22.0

Validation Checkpoint - Verify Agent Availability

Execution

Open the **Agents** page from the left navigation menu.

Expected Output / Outcome

The **MathTutorAgent-1** agent is visible in the Agents list.

Validation Checkpoint - Verify Tool Execution

Execution

In the Playground chat, review the generated response and code execution details.

Expected Output / Outcome

The agent displays:

- The Python code used to solve the equation.
- The code execution output.
- The final answer (x = 5).

This confirms that the **Code Interpreter** tool is working correctly.

Cleanup Steps

Step 1 - Delete the Agent

Execution

1. Open the **Agents** page from the left navigation menu.
2. Select **MathTutorAgent-1** from the agents list.
3. Click **Delete** and confirm the deletion.

Expected Output / Outcome

The agent is deleted successfully and no longer appears in the agents list.

Step 2 - Clear the Playground Conversation

Execution

Clear the current conversation or start a new chat session from the Playground.

Expected Output / Outcome

The current conversation history is cleared and a new session is available for testing.

Step 3 - Sign Out

Execution

Click the profile icon in the top-right corner and select **Sign out**.

Expected Output / Outcome

The user is signed out and redirected to the sign-in page.

Final Lab Outcome

You successfully accessed the Microsoft Foundry portal, opened the Azure AI Foundry project workspace, verified the **gpt-5.4-mini** deployment, created an autonomous AI agent, enabled the **Code Interpreter** tool, and validated code execution through the Agent Playground.